

A Home page and validated workflow

HOMEANALYSISPREDICTHELP

DECIDInBRG

MALER

Version: 1.0

Welcome to MALER

A web server to build, evaluate, and apply machine learning models for numerical biological and biomedical data analysis.

What is MALER?

Machine learning (ML) methods have accelerated discoveries in biological and biomedical research, especially in the analysis of big omics data. However, biologists often lack knowledge of ML theories and programming skills to easily apply ML in their research. The lack of a one-stop web server to build, evaluate, and compare the performance of different ML models for biological data has been a major obstacle for biologists to apply ML widely.

We developed **MALER** (Machine Learning Evaluation and Reporting server), a guided web application that helps biologists apply ML models to biological and biomedical data without writing code. MALER provides modules for (i) binary and (ii) multi-class classification, and (iii) regression and (iv) survival analysis that allow users to build, quantitatively evaluate, and compare multiple ML models with task-appropriate metrics. The validated workflow fits imputation, scaling, feature ranking or reduction, and model tuning within training folds; evaluates the declared test partition only after model development; and supports identifier-aligned external prediction with a signed, downloadable model bundle.

In the **ANALYSIS** page, users can upload their omics profile dataset and the corresponding labels to build and compare their custom ML models with step by step guidance. In the **PREDICT** page, users can upload their ML model and a omics profile with unknown labels to predict outcomes. You can find detailed instructions on how to use MALER on the **HELP** page.

MALER Workflow

Choose an analysis task and upload data data

BinaryMulticlassRegressionSurvival

Development data

Training and model selection

Leakage-controlled development

Fold local pipeline

refit per fold: imputation, scaling, reduction, model

Nested cross validation

Inner tuning, outer performance estimation

Lock and fit the selected pipeline

development data only

Export signed .maler model

filled pipeline and expected feature identifiers

External data

identifier-aligned features

External prediction without retraining

predicted outcomes or risk scores

Hold-out test

Model and evaluation

No preprocessing

No feature selection

No model tuning

Evaluate once

after all models are locked

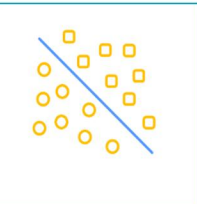
News & Updates

MALER 1.0 was released.

[2023-12-18]

B Analysis task selection

Step 1: Please select an analysis task.



Binary classification

The process of dividing new observations into distinct two categories based on the features of a set of input data.



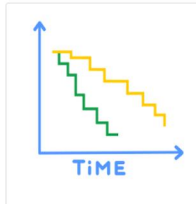
Multiclass classification

The process of dividing new observations into distinct at least three categories based on the features of a set of input data.



Regression

The process of exploring the relationships between dependent variables (labels) and a series of independent variables (features).



Survival

The process of establishing the connection between covariates (features) and the time of events (labels) to predict the risk of future events.

C Upload, privacy and access controls

Step 2: Please upload the data file.

Choose data to upload:

File:

Upload

☒ Upload Your Own Data

☐ Use Example/Template Data

File:

Upload a CSV file with features in rows and samples in columns. The file must be no larger than 200MB. Example/template files are provided to illustrate the required input format.

Template:

Classification and regression files use the first three rows for sample name, label, and train/test set. Survival files use the first four rows for sample name, status, time, and train/test set. Example files can be downloaded below.

Project access:

MALER generates a random project suffix and a separate access token. Keep the result link private; the project ID alone is not sufficient to retrieve data or results.

☐ I confirm that this upload contains no directly identifying information and that I am authorized to process these data.

D Feature-scaling guidance

Step 3: Please select a feature scaling method. (Optional)

Feature Scaling Methods:

☐ MinMax

☒ Z-Score

☐ MaxAbs

☐ None

Z-Score is the recommended default for fair comparison of scale-sensitive SVM, KNN and penalized linear models. Tree-based models are generally less scale sensitive. The setting can be changed, including to no scaling; doing so for a scale-sensitive model may make comparisons unfair. Scaling is fitted inside every training fold, and the selected setting is recorded with the result.

E Feature ranking and subset search

Step 4: Please select feature ranking and subset search methods.

Feature Ranking Methods:

☒ ANOVA

☐ MRMR

ANOVA is suitable for fast univariate ranking in classification or regression tasks. MRMR considers relevance and redundancy and is useful when many correlated features are expected. Cox ranking is used for survival analysis.

Subset Search Methods:

☒ TopK

☐ FSS

☐ BSS

☐ PCA

☐ No reduction

TopK tunes the retained feature count. FSS and BSS use task-specific sequential selection after a fold-local prescreen capped at 50 candidates. PCA is unsupervised and returns components rather than a gene signature. Every option is refit independently inside each training fold.

F Analysis strategy and validation

Step 5: Please choose an analysis strategy.

One-Click Analysis

Customized Analysis

One-Click Analysis:

The model is built with the default parameters and the data is analyzed using all the models provided.

Customized Analysis:

Users can choose to specify models and parameters to personalize the data.

Validation:

The default design uses repeated 5-fold outer validation (2 repeats) and 3-fold inner tuning. Classification is optimized by balanced accuracy, regression by R², and survival by concordance index. Splits are reduced automatically only when the cohort is too small.

Complete parameter record:

All estimator classes, defaults, scaling requirements and searched values are available in the versioned [machine-readable method registry](#).

G Signed model and external prediction

Step2: Please upload the files and models.

Choose data to upload:

☒ Upload Your Own Data

☐ Choose Example Data

Model: Upload .maler

Data: Upload CSV

Model: Upload a signed .maler model bundle downloaded from MALER Analysis.

File: Upload a CSV file with features in rows and samples in columns, no larger than 200 MB.

Privacy: Do not upload direct identifiers or sensitive patient data. Access tokens expire after 24 hours by default; cached files are physically removed when the operator runs the cleanup job.

☐ I confirm that the uploaded data are de-identified and that I have consent, ethics approval, and institutional authorization where required.